NC Spot Drill >>

NC Spot Drill with indexable carbide insert.

High efficiency! Low cost!

CNC lathes, CNC turning centers and machining centers.

1 NC Spot Drill

Features

- **Spotting produces better hole position and geometrically uniform holes**
- **Available shank diameter- Ø5, Ø6, Ø8, Ø10, Ø12, Ø16, Ø20, Ø25mm, Ø3/8'', Ø1/2'', Ø5/8'', Ø1/4", Ø3/4'', M5, M6 and M8.**
- **One tool will perform multiple applications**
  - Long tool life.
  - Each insert has 2 or 4 cutting edges.
  - Suitable for spotting, chamfering, grooving and engraving.
  - 60° / 82° / 90° / 100° / 120° / 142° angle for different applications.
  - Increase cutting speed with coated carbide inserts.



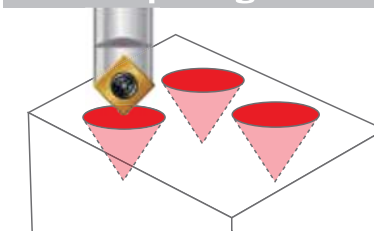


▲Machining Center

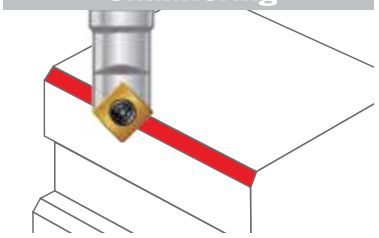
a Engraving
b Spotting
c Chamfering
d Grooving

ALL IN ONE!!

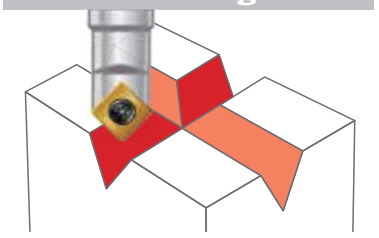
Spotting



Chamfering



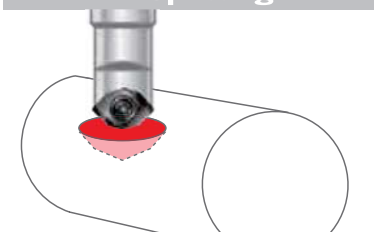
Grooving



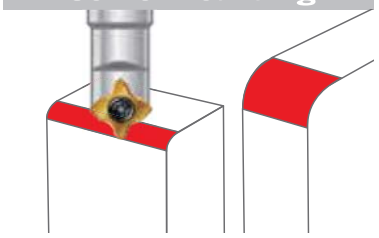
Engraving



W Spotting



Corner Rounding



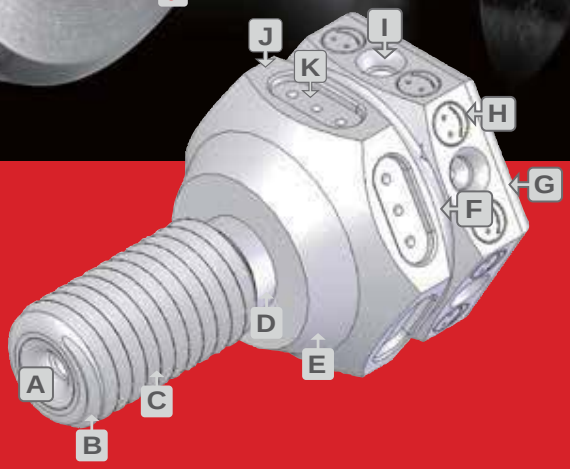
▲CNC Lathes

- a External and internal chamfering
- b Grooving
- c Centering
- d Facing



1

NC Spot Dril



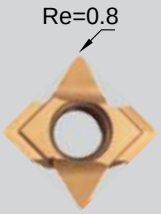
Multifunctional:

A I	Center Drilling	B G	Corner rounding
C	Thread turning	D	Grooving
E	Taper turning	F	V-grooving
H	Engraving	J	Face milling
K	Drilling & milling a groove		

* Some features produced with a special insert

60° N9MT11T3P60

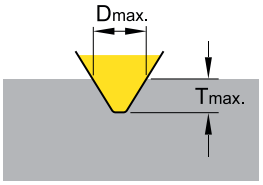


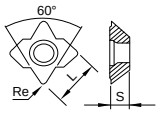


Inserts >>

• Fully ground spotting insert, for 60 degree spotting and engraving.

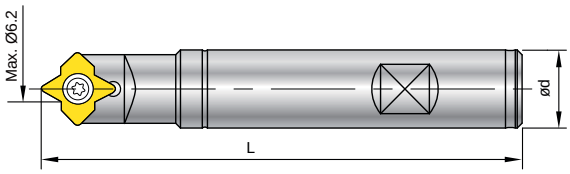
NC40: • Universal grade for all unhardened steel and cast iron.
• Each insert has 2 cutting edges.



Code	Parts No.	Coating	Grade		Dimensions			Dmax.	Tmax.
					L	S	Re		
014204	N9MT11T3P60-NC40	TiN	P35		11	3.97	0.8	6.2	4

Holder >>

• A single cutting edge design creates higher precision and position when spotting.
• Applications: For spotting, engraving, small grooving on milling machines, machining centers.

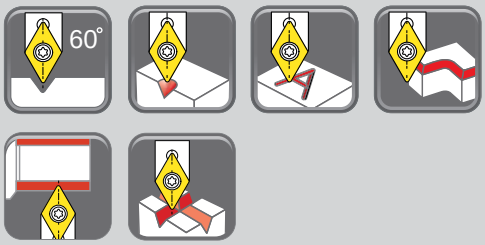


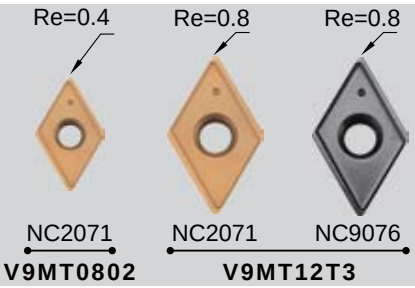
Ø16



Code	Parts No.	Ød	L	Screw	Key
604002	00-99616-14-12	12	100	 NS-35080 2.5 Nm	 NK-T15
604004	00-99616-14	16	100		

V9MT0802 / V9MT12T3



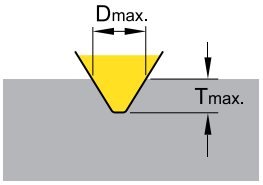


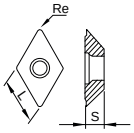
Inserts >>

- 60 degree indexable spotting insert, Dmax 13mm.
- Special geometry with supporting edges for using in high speed machining.
- Excellent tool for grooving. Saving machining time!

NC2071:
- Universal grade for all unhardened steel and cast iron.
- Each insert has 2 cutting edges.

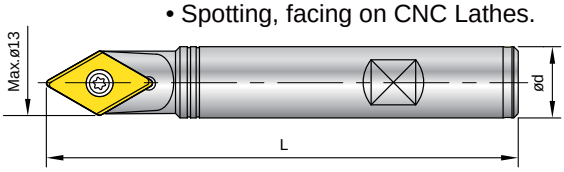
NC9076:
- For non-ferrous material such as aluminum, al-alloy, titanium brass, copper and long cutting chip metal.
- Produces excellent surface finish on non-ferrous metal.
- Each insert has 2 cutting edges.



Code	Parts No.		Coating	Grade		Dimensions L	Dimensions S	Dimensions Re	Dmax.	Tmax.
019201	V9MT0802CT	NC2071	TiN	K20F		8	2.38	0.4	9	7.3
015201	V9MT12T3CT	NC2071	TiN	K20F		12.7	3.97	0.8	13	10.3
015202		NC9076	DLC	K20F						

Holder >>

- A single cutting edge creates higher precision and position when spotting.
- Applications: • Spotting, engraving, grooving and chamfering on milling machines, machining centers.
  • Spotting, facing on CNC Lathes.

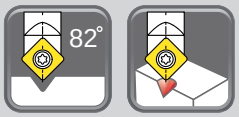


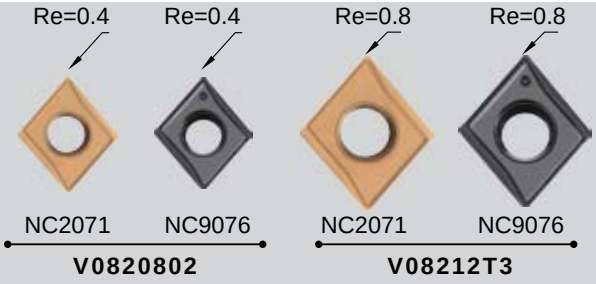


Code	Parts No.	Ød	L	Insert Type	Screw	Key
609001	00-99616-09V (Cylindrical shank)	8	60	V9MT08	*NS-25045 0.9 Nm	NK-T7
605001	00-99616-13V	16	100	V9MT12	NS-35080 2.5 Nm	NK-T15
615001	00-99616-13V-5/8	5/8”	100			

*Torque screwdriver is recommended, see page 6-4.

82° V0820802 / V08212T3



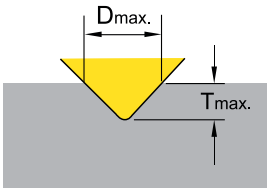


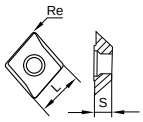
Inserts >>

- 82 degree indexable spotting insert, Dmax 14mm (0.551")
- Match the geometry of Amercian standard flat head screw hole.
- Special geometry with supporting edges for high speed machining.

NC2071:
- Universal grade for all unhardened steel and cast iron.
- Each insert has 2 cutting edges.

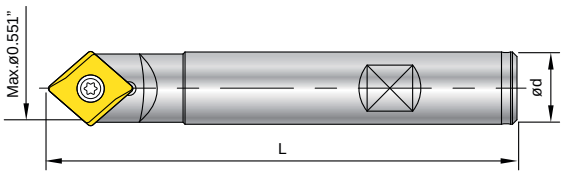
NC9076:
- For non-ferrous material such as aluminum, al-alloy, titanium brass, copper and long cutting chip metal.
- Produces excellent surface finish on non-ferrous metal.
- Each insert has 2 cutting edges.



Code	Parts No.		Coating	Grade		Dimensions			Dmax.	Tmax.
						L	S	Re		
0108201	V0820802	NC2071	TiN	K20F		8	2.38	0.4	9 (0.354")	4.8 (0.189")
0108202		NC9076	DLC							
0108211	V08212T3	NC2071	TiN	K20F		12.7	3.97	0.8	14 (0.551")	7.5 (0.295")
0108212		NC9076	DLC							

Holder >>

- Special cutting edge design gives higher precision and position when spotting.
- Applications : • Spotting, engraving, grooving and chamfering on milling machines, machining centers.
  • Spotting, facing on CNC Lathes.



Ø3/8"



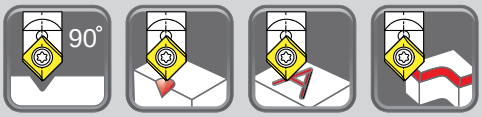
Ø5/8"

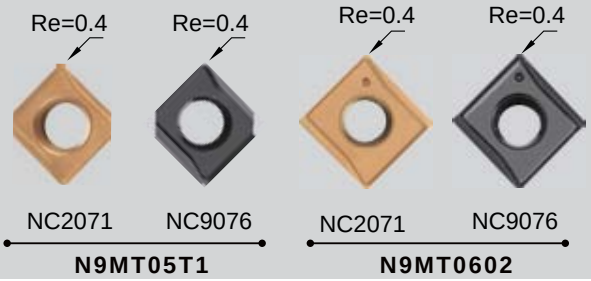


Code	Parts No.	Ød	L	Insert Type	Screw	Key
693001	00-99619-V082-3/8	3/8"	90	V0820802	NS-30055 2.0 Nm	NK-T8
693002	00-99619-V082-5/8	5/8"	100	V08212T3	NS-35080 2.5 Nm	NK-T15

N9MT05T1 / N9MT0602

90°



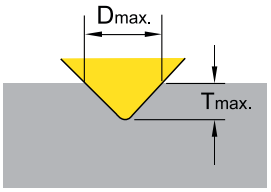


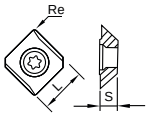
Inserts >>

- Mini spotting drill with indexable insert, low cutting power required.
- Especially good for Swiss type automatic lathes and CNC lathes.

NC2071:
- Universal grade for all unhardened steel and cast iron.
- Geometry with supporting edges to stabilize the cutting condition on low power machine.
- Each insert has 2 cutting edges.

NC9076:
- For non-ferrous material such as aluminum, titanium, brass, copper and stainless steel.
- Produces excellent surface finish on non-ferrous metal.
- Each insert has 2 cutting edges.



Code	Parts No.		Coating	Grade		Dimensions L	Dimensions S	Dimensions Re	Dmax.	Tmax.
011201	N9MT05T1CT	NC2071	TiN	K20F		5	1.8	0.4	6	2.8
011202		NC9076	DLC	K20F						
012201	N9MT0602CT	NC2071	TiN	K20F		6.35	2.38	0.4	8	3.8
012202		NC9076	DLC	K20F						

Holder >>

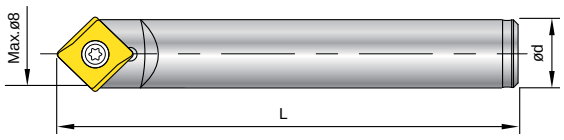
- Smallest indexable spotting drill holder.
- Single cutting edge design gives higher precision when spotting.
- Applications : • Spotting, engraving, and chamfering on milling machines, machining centers.
  • Spotting, facing on CNC Lathes.









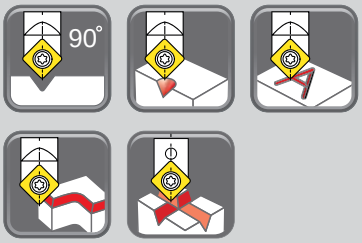


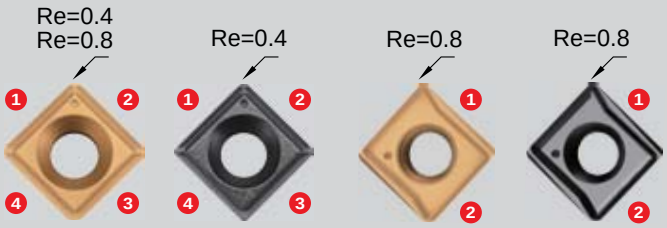
Code	Parts No.	Ød	L	Insert Type	Screw	Key
601001	00-99616-06-6	6	35	N9MT05	*NS-20036 0.6 Nm	NK-T6
601002	00-99616-06-5	5	35			
601003	00-99616-06-6L	6	60			
602001	00-99616-08-8	8	60	N9MT06	*NS-22044 0.9 Nm	NK-T7

Note:601003 is carbide shank holder.

*Torque screwdriver is recommended, see page 6-4.

90° N9MT0802





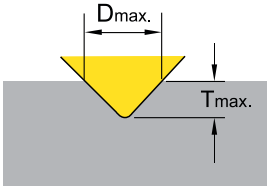
Inserts >>

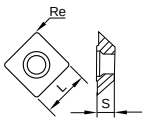
NC40:
- General purpose, universal grade for all unhardened steel.
- Each insert has 4 cutting edges.

NC10:
- High positive angle and fully ground cutting edge and relief angle.
- Universal grade for non-ferrous metal, cast iron and stainless steel.
- Each insert has 4 cutting edges.

H-NC40:
- Best choice for spotting application.
- Special geometry with supporting edges for use in high speed machining.
- Universal grade for all kind of steel and cast iron.
- Each insert has 2 cutting edges.

H-NC9076:
- High positive geometry and sharp edge.
- For non-ferrous material such as aluminum, titanium, brass, copper and long cutting chip metal.
- Produces excellent surface finish on non-ferrous metal.
- Each insert has 2 cutting edges.

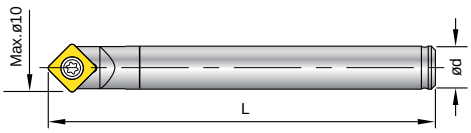


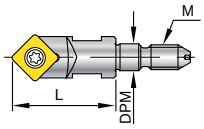
Code	Parts No.		Coating	Grade		Dimensions L	S	Re	Dmax.	Tmax.
013401	N9MT080208CT	NC40	TiN	K20F		8.31	2.38	0.8	10	4.5
013402	N9MT080204CT	NC40	TiN	K20F		8.31	2.38	0.4		
013403		NC10	TiAlN			8.31	2.38	0.4		
013201	N9MT0802CT2T	H-NC40	TiN	K20F		8.31	2.38	0.8		
013202		H-NC9076	DLC			8.31	2.38	0.8		

Holder >>

- Single cutting edge design gives higher precision when spotting.
- Applications : • Spotting, engraving, grooving and chamfering on milling machines, machining centers.
  • Spotting, facing, turning on CNC Lathes.











Code	Parts No.	Ød	L	M	DPM	Screw	Key
603001	00-99616-10	10	90	-	-	 NS-30055 2.0 Nm	 NK-T8
603003	00-99616-10-SL10 (Weldon)	10	90	-	-		
613001	00-99616-3/8	3/8”	90	-	-		
623001	00-99616-10-M5	-	25	M5xP0.8	5.5		
623002	00-99616-10-M6	-	25	M6xP1.0	6.5		

Note: • Balanced type holder is on request.
• Nine9 extension bar for M5, M6 screw fit holder, see page 6-3.

N9MT0802

90°



Single Set >>

• User friendly, each set is fitted with one complimentary insert.

Code	Parts No.	Ød	Total Length	Insert fitted	Dmax.	Tmax.
603101-3401	00-99616-10-02S	10	90	N9MT080208CT-NC40	10	4.5
603101-3403	00-99616-10-02SAL	10	90	N9MT080204CT-NC10	10	4.5

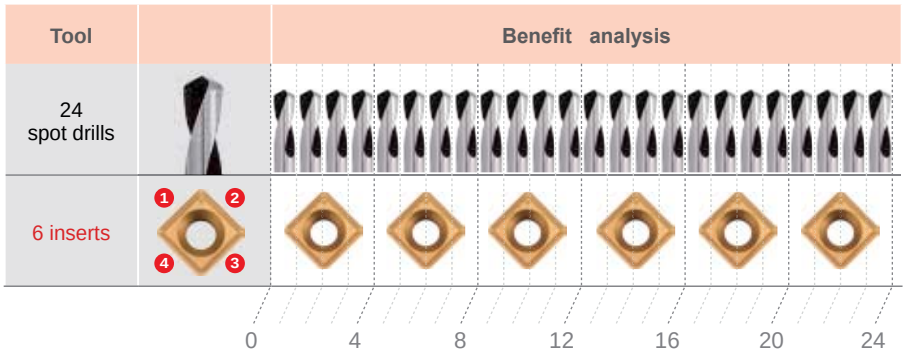
Starter Package >>

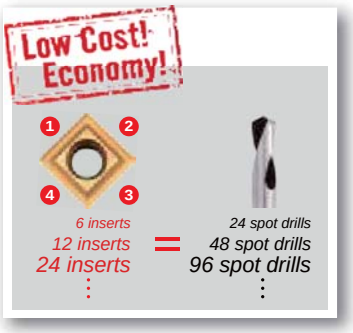
• Selected package for starter who wants to try NC Spot Drill.
• Included one insert on tool holder and five inserts in the pocket.
• Total 6 inserts are equal to 24 spot drills.

Code	Parts No.	Ød	Insert included	Content
603201-3401	00-99616-10-ME6	10	N9MT080208CT-NC40	1 tool holder + 6 inserts + 1 key
603201-3403	00-99616-10-ME6AL	10	N9MT080204CT-NC10	
613201-3401	00-99616-10-IN6	3/8”	N9MT080208CT-NC40	
613201-3403	00-99616-10-IN6AL	3/8”	N9MT080204CT-NC10	



Comparison >>



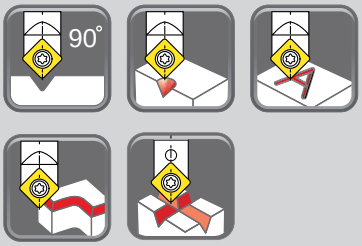


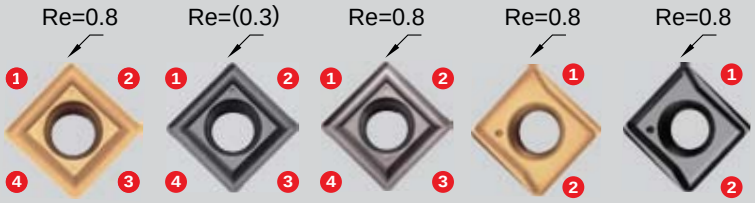
Note: N9MT080201W Engraving , see page 1-47.



60°-NC40 NC40 NC10

90° N9MT11T3





Inserts >>

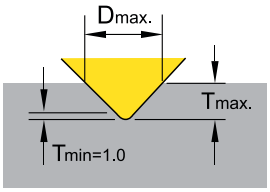
NC40:
- Wiper design, universal grade for all unhardened steel.
- Each insert has 4 cutting edges.

NC10:
- High positive angle and fully ground cutting edge and relief angle.
- Universal grade for non-ferrous metal, cast iron and stainless steel.
- Each insert has 4 cutting edges.

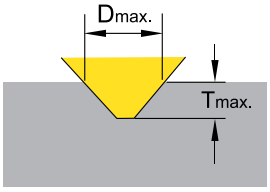
NC60:
- Wiper design cermet insert, for hardened steel up to 56 HRC.
- Each insert has 4 cutting edges.

H-NC40:
- Best choice for spotting application.
- Special geometry with supporting edges for use in high speed machining.
- Universal grade for all kind of steel and cast iron.
- Each insert has 2 cutting edges.

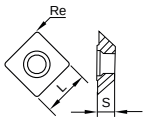
H-NC9076:
- High positive geometry and sharp edge.
- For non-ferrous material such as aluminum, titanium, brass, copper and long cutting chip metal.
- Produces excellent surface finish on non-ferrous metal.
- Each insert has 2 cutting edges.



NC40 / Wiper design / NC60

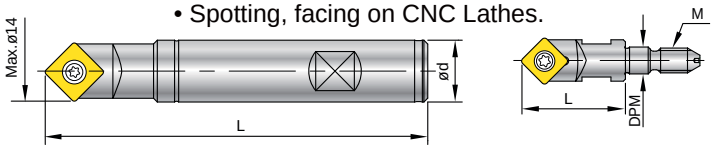


Other grade

Code	Parts No.		Coating	Grade		Dimensions L	Dimensions S	Dimensions Re	Dmax.	Tmax.
014401	N9MT11T3CT	NC40	TiN	P35		11.11	3.97	0.8	14	7
014402		NC10	TiAlN	K10F				(0.3)		
014403		NC60	CERMET					0.8		
014202	N9MT11T3CT2T	H-NC40	TiN	K20F				0.8		
014203		H-NC9076	DLC	K20F				0.8		

Holder >>

- Single cutting edge design gives higher precision when spotting.
- Applications : • Spotting, engraving, grooving and chamfering on milling machines, machining centers.
  • Spotting, facing on CNC Lathes.







Code	Parts No.	Ød	L	M	DPM	Screw	Key
604002	00-99616-14-12	12	100	-	-	NS-35080 2.5 Nm	NK-T15
604004	00-99616-14	16	100	-	-		
604007	00-99616-14-150L	16	150	-	-		
604009	00-99616-14-220L	20	220	-	-		
614001	00-99616-14-1/2	1/2”	100	-	-		
614002	00-99616-14-5/8	5/8”	100	-	-		
624001	00-99616-14-M8	-	30	M8xP1.25	8.5		

Note: • Balanced type holder is on request.
• Nine9 extension bar for M8 screw fit holder, see page 6-3.



Single Set >>

• User friendly, each set is fitted with one complimentary insert.

Code	Parts No.	Ød	Total Length	Insert fitted	Dmax.	Tmax.
604104-4401	00-99616-14-02S	16	100	N9MT11T3CT-NC40	14	7
604104-4402	00-99616-14-02SAL			N9MT11T3CT-NC10	14	7
614102-4401	00-99616-14-5/8-02S	5/8”	100	N9MT11T3CT-NC40	0.551”	0.276”
614102-4402	00-99616-14-5/8-02SAL			N9MT11T3CT-NC10	0.551”	0.276”

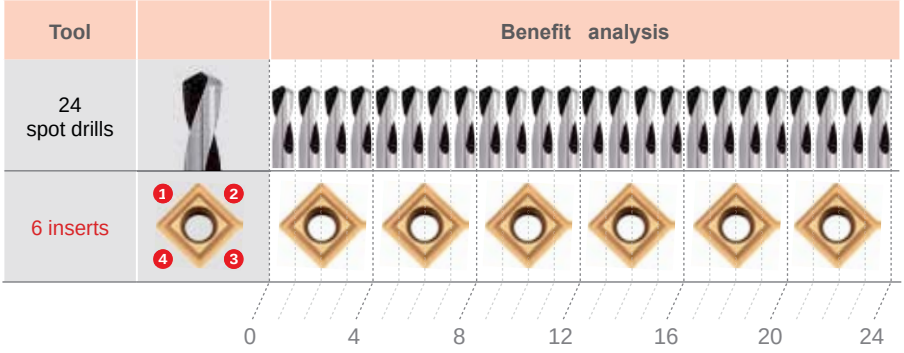
Starter Package >>

• Selected package for starter who wants to try NC Spot Drill.
• Included one insert on tool holder and five inserts in the pocket.
• Total 6 inserts are equal to 24 spot drills.

Code	Parts No.	Ød	Insert included	Content
604204-4401	00-99616-14-ME6	16	N9MT11T3CT-NC40	1 tool holder + 6 inserts + 1 key
604204-4402	00-99616-14-ME6AL		N9MT11T3CT-NC10	
614202-4401	00-99616-14-IN6	5/8”	N9MT11T3CT-NC40	
614202-4402	00-99616-14-IN6AL		N9MT11T3CT-NC10	



Comparison >>





90° N9MT1704



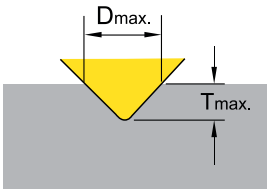


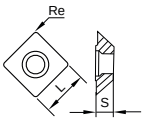
NC2071

Inserts >>

• 90 degree indexable spot drill insert, Dmax 22mm.

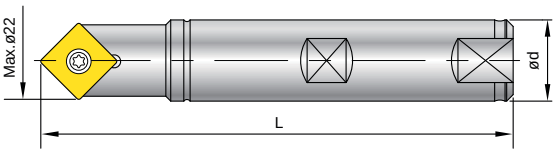
NC2071 : • High positive geometry,
fully ground cutting edge and relief angle.
• Universal grade for all unhardened steel and cast iron.
• Each insert has 2 cutting edges.



Code	Parts No.	Coating	Grade		Dimensions			Dmax.	Tmax.
					L	S	Re		
016201	N9MT1704CT-NC2071	TiN	K20F		17	4.76	1.2	22	10.4

Holder >>

• Single cutting edge design gives high precision when spotting.
• Applications : • Spotting, engraving, grooving and chamfering on milling machines, machining centers.
• Spotting, facing on CNC Lathes.



Ø20



Code	Parts No.	Ød	L	Screw	Key
606001	00-99616-22	20	100	NS-50125 5.5 Nm	NK-T20
606002	00-99616-22-25	25	150		

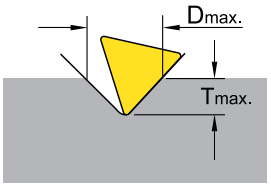
N9MT220408





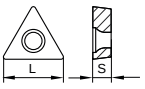
NC40

Inserts >>



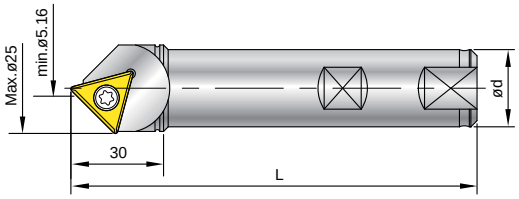
- For spotting diameter up to 25mm.
- Fully ground cutting edge and relief angle.

NC40:
- Universal grade for carbon steel, alloy steel and cast iron.
- Each insert has 3 cutting edges.

Code	Parts No.	Coating	Grade		Dimensions			Dmax.	Tmax.
					L	S	Re		
017301	N9MT220408CT-NC40	TiN	P35		20.83	4.76	---	25	12.2

Holder >>

- Large spotting diameter with indexable insert.
- Single cutting edge design gives high precision when spotting.
- Applications : spotting and chamfering on milling machine, machining centers.



Ø25



Code	Parts No.	Ød	L	Screw	Key
607001	00-99616-25-CT28	25	120	 NS-40100 3.5 Nm	 NK-T15
617001	00-99616-1-CT28	1”	120		

N9MT11T3CT2T-H

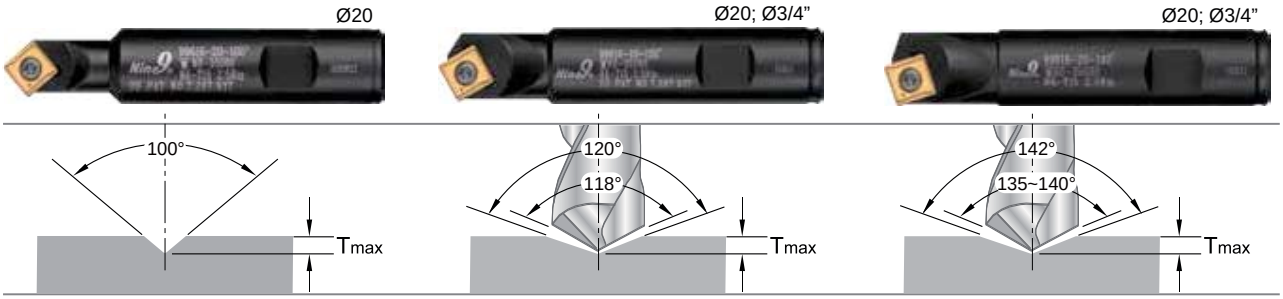




H-NC40

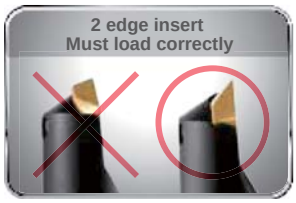


H-NC9076



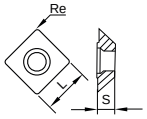
- For aircraft 100° normal rivet hole and screw hole.
- For spotting before drilling by 118° point angle drill.
- 60° chamfering.
- For spotting before drilling by 135°~140° point angle high performance drill.

Inserts >>



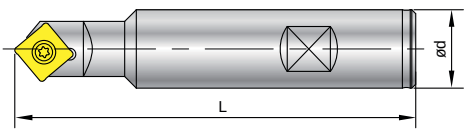
H-NC40:
- Universal grade for all kind of steel and cast iron.
- Each insert has 2 cutting edges.

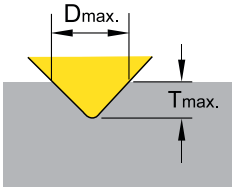
H-NC9076:
- High positive geometry and sharp edge.
- For non-ferrous material such as aluminum, titanium, brass, copper and long cutting chip metal.
- Produces excellent surface finish when chamfering non-ferrous metal.
- Each insert has 2 cutting edges.

Code	Parts No.		Coating	Grade		Dimensions		
						L	S	Re
014202	N9MT11T3CT2T	H-NC40	TiN	K20F		11	3.97	0.8
014203		H-NC9076	DLC					

Holder >>

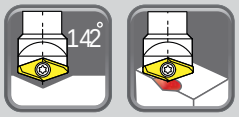
- Indexable insert spotting drill holders for 100°/120°/142° spotting.
- Spotting produces better hole position and geometrically uniform holes.
- Increase tool life of the next drilling operation.



Code	Parts No.	Angle	Ød	L	Screw / Key	Dmax.	Tmax.	
604011	00-99616-20-100	100°	20	100	 NS-35080 2.5 Nm	16	6.3	
604013	00-99616-20-120	120°	20	100		17	4.76	
614003	00-99616-3/4-120	120°	3/4"	100	 NK-T15	0.669"	0.187"	
604014	00-99616-20-142	142°	20	100		18.5	3.16	
614004	00-99616-3/4-142	142°	3/4"	100		0.728"	0.124"	

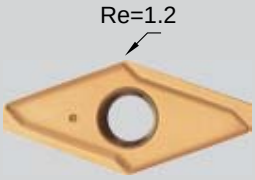
V14208 / V14216

142°





V1420803-NC2071

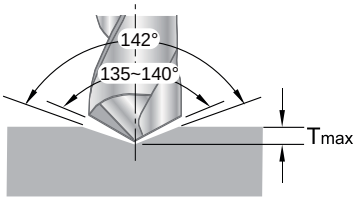


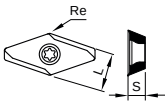
V1421604-NC2071

Inserts >>

- For spotting before drilling by 135° - 140° point angle high performance drill.
- 142 degree indexable spotting drills. Dmax 32mm.

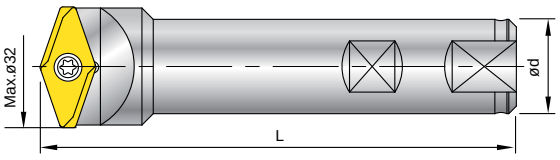
NC2071:
- High positive geometry, fully ground cutting edge and relief angle.
- Universal grade for all unhardened steel and cast iron.
- Each insert has 2 cutting edges.



Code	Parts No.	Coating	Grade		Dimensions			Dmax.	Tmax.
					L	S	Re		
0114201	V1420803-NC2071	TiN	K20F		8	2.38	0.8	16	2.8
0114211	V1421604-NC2071				14	4.76	1.2	32	5.5

Holder >>

- Using spotting first may increase higher speed and feed rate of the after drills.
- Extend your drill life with 142° spotting. Reduce your drilling cost.
- Higher accuracy of positioning and diameter tolerance !



Ø16



Ø25

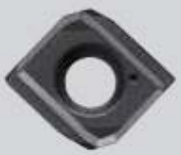


Code	Parts No.	Ød	L	Insert Type	Screw	Key
696001	00-99619-V142-16	16	100	V1420803	NS-30072 2.0 Nm	NK-T9
696002	00-99619-V142-32	25	120	V1421604	NS-50125 5.5 Nm	NK-T20

145°
+
90°

WSP Spotting New Geometry of Spotting Tool

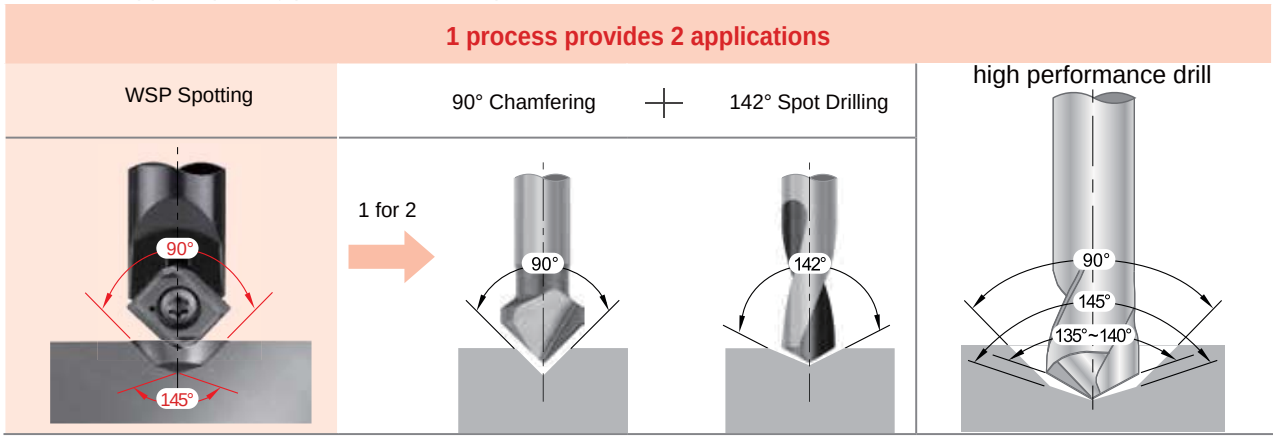




NC2033

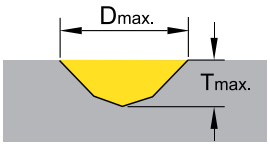
Combined spotting and chamfering 145° + 90° >>

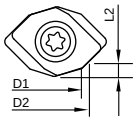
- Reduces process to one operation. Shorten cycle time.
- Use to spot prior to drilling with high performance drills for higher accuracy of hole position.
- Good support spotting process for round parts.



Inserts >>

NC2033:
- Fully ground cutting edge and relief angle.
- Universal grade for steel and cast iron.
- Each insert has 2 cutting edges.



Code	Parts No.	Coating	Grade		Thread Size	*D1±0.05	D2	L2	Dmax.	Tmax.
013203	N9MT0802M04C-NC2033	TiAlN	K20F		M4x0.7	3.30	4.20	0.93	8	2.83
013204	N9MT0802M05C-NC2033				M5x0.8	4.20	5.25	1.14		2.52
013205	N9MT0802M06C-NC2033				M6x1.0	5.00	6.30	1.39		2.24
014219	N9MT11T3M08C-NC2033	TiAlN	K20F		M8x1.25	6.80	8.40	1.81	13	4.11
014220	N9MT11T3M10C-NC2033				M10x1.5	8.50	10.50	2.28		3.53
014221	N9MT11T3UNC25-NC2033	TiAlN	K20F		1/4-20 UNC	5.08	6.70	1.55	13	4.70
014222	N9MT11T3UNC31-NC2033				5/16-18 UNC	6.53	8.40	1.90		4.20
014223	N9MT11T3UNC38-NC2033				3/8-16 UNC	7.94	10.00	2.22		3.72
016205	N9MT1704M12C-NC2033	TiAlN	K20F		M12x1.75	10.25	12.60	2.91	20	6.61
016206	N9MT1704M14C-NC2033				M14x2.0	12.00	14.70	3.22		5.87
016207	N9MT1704M16C-NC2033				M16x2.0	14.00	16.80	3.51		5.11

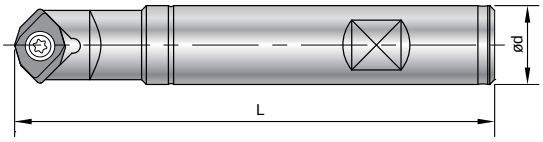
Note: * D1 refer to the Tap Pre-drilling sizes.
* Technical information, please refer to page 1-25.

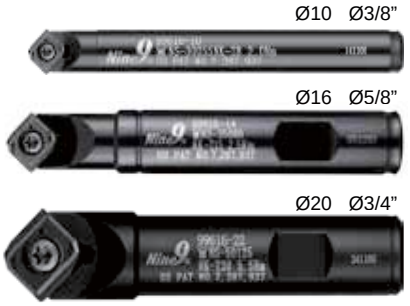
WSP Spotting New Geometry of Spotting Tool



Holder >>

- Utilizes standard ***NC Spot Drill*** holders.
- Holders and inserts are interchangeable.



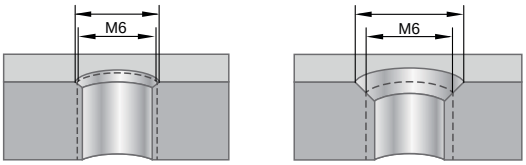


Code	Parts No.	Ød	L	Insert Type	Thread Size	Screw	Key
603001	00-99616-10	10	89.08±0.29	N9MT0802	M4~M6	NS-30055 2.0Nm	NK-T8
613001	00-99616-3/8	3/8”					
604004	00-99616-14	16	97.55±0.55	N9MT11T3	M8~M10	NS-35080 2.5Nm	NK-T15
614002	00-99616-14-5/8	5/8”			1/4~3/8 UNC		
606001	00-99616-22	20	96.24±0.64	N9MT1704	M12~M16	NS-50125 5.5Nm	NK-T20
616001	00-99616-22-3/4	3/4"					

Example >>

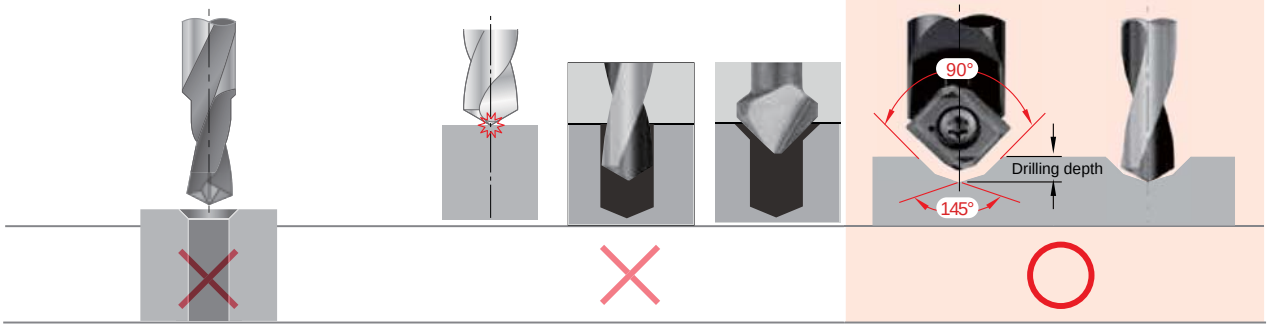
- The recommended chamfering is 5% of the nominal diameter of the thread, for example 6.3 mm for M6 thread.
- If you need larger chamfer, it can be calculated the required depth of spotting. (see page 1-27)

Drilled hole for M6	
M6+5% Chamfering	M6+1mm Chamfering



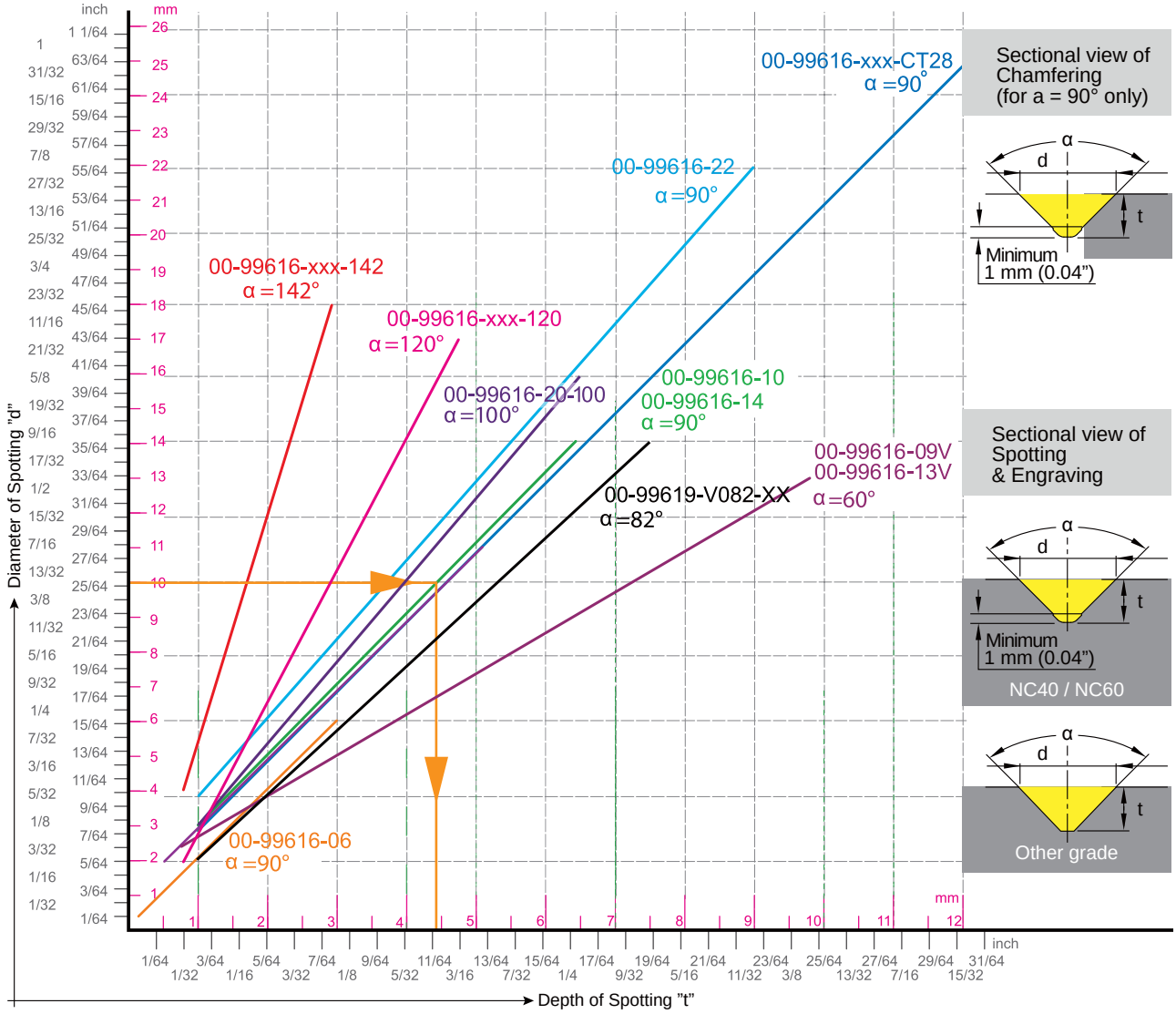
Comparison >>

Step Drill	Drill + Spotting	WSP Spotting + Drill
• Tool cost is high • Shorter tool life • Can't drill directly from solid on round parts. Bad position accuracy.	• Longer drilling time • Guided at the weakest corner of drill • Shorter tool life	• Shorter drilling time • Guided at the strongest corner of drill • Longer tool life • Good position accuracy



Cutting Data

Diameter / Depth Chart and Speed / Feed Rate Calculation of NC Spot Drill



Instruction of Use >>

1. From Spot diameter "d" to get drill depth " t ".
2. Point angle "α" is determined by which tool holder you use.
3. From "d" draw a horizontal line to get intersection of the line by point angle "α".
4. From the intersection draw a vertical line to the bottom to have depth of spotting " t ". " t " is the drill depth of the NC program.
5. The sectional view of spotting will depend on the shape of insert, NC40 and other grades of inserts have different sectional view.
6. For chamfering, do not use tip of insert, 1mm(0.04") minimum clearance is required for a smooth surface finish.

Calculate spindle speed and feed rate >>

1. Using your "d" value and cutting speed Vc from the data sheet, calculate spindle speed "S"(RPM).
2. " F" feed rate per minute F = f x S = RPM x IPR

Metric		Inch	
$S = \frac{Vc \times 1000}{\pi \times d}$	**d =** diameter -mm	$S = \frac{(3.82 \times SFM)}{d}$	**d =** diameter-inch
	S = Spindle Speed -r.p.m.		**S =** Spindle Speed-r.p.m.
	Vc = Cutting Speed -m/min.		**SFM =** Surface Speed-ft./min. Vc (m/min.) x 3.28
F = S x f	**f =** mm/rev.	**F = f x S**	**f = IPR =** inch/rev.
	F = mm/min.		**F =** inch/min.

Cutting Data

N9MT-CT >> Insert Multi-function

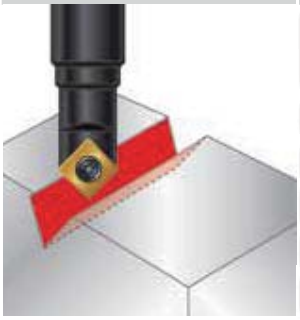
Determine spindle speed and feed rate:

- Choose spotting depth to decide spotting diameter according to the Diameter/Depth chart on page 1-25.
- The spindle speed should be calculated by the maximum diameter of spotting, chamfering and grooving.

Spotting	Work Material	Vc (m/min)	f (mm/rev.)	Grade of Insert
	Carbon steel	150~250	0.05~0.10	NC40, NC2071
	Alloy steel	100~200	0.04~0.08	NC40, NC2071
	Stainless steel	65~125	0.03~0.06	NC10, NC60, H-NC40, NC2071
	Casting iron	80-150	0.05~0.10	H-NC40, NC10, NC2071
	Non-ferrous metal (Al, Cu)	150~300	0.05~0.10	NC10, NC9076, NC2071
	Ti, Ti-alloy	40~80	0.03~0.08	NC9076
	Hardened steel 40~56 HRC	30~60	0.03~0.08	NC60

* For technical construction reasons, the insert is not located on the center of the holder.
* Inserts with supporting edges can increase feed rate 50%.

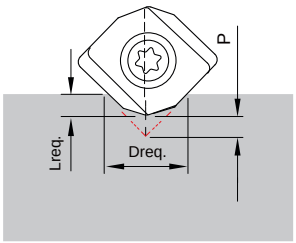
Chamfering	Work Material	Vc (m/min)	f (mm/rev.)	Grade of Insert
	Carbon steel	150~320	0.15~0.24	NC40, NC2071
	Alloy steel	100~250	0.12~0.20	NC40, NC2071
	Stainless steel	65~125	0.1~0.20	NC10, NC60, H-NC40, NC2071
	Casting iron	150-250	0.15~0.25	H-NC40, NC10, NC2071
	Non-ferrous metal (Al, Cu)	150~320	0.15~0.25	NC10, NC9076, NC2071
	Ti, Ti-alloy	40~80	0.03~0.08	NC9076
	Hardened steel 40~56 HRC	30~60	0.03~0.08	NC60

Grooving	Work Material	Vc (m/min)	f (mm/rev.)	Grade of Insert
	Carbon steel	150~250	0.05~0.10	NC40, NC2071
	Alloy steel	100~200	0.04~0.08	NC40, NC2071
	Stainless steel	65~125	0.03~0.06	NC10, NC60, H-NC40, NC2071
	Casting iron	80~150	0.05~0.08	H-NC40, NC10, NC2071
	Non-ferrous metal (Al, Cu)	150~320	0.05~0.08	NC10, NC9076, NC2071
	Ti, Ti-alloy	40~80	0.03~0.08	NC9076
	Hardened steel 40~56 HRC	30~60	0.03~0.08	NC60

Cutting Data

WSP Spotting >> 145°+90° W Spotting

WSP spotting	Formula



Lreq. = Dreq. x 0.5 - P

P =	distance of theoretical intersection point to tip of insert.
0.5 =	fixed factor for calculation
Lreq. =	required drilling depth
Dreq. =	required diameter

	M4	M5	M6	M8	M10	M12	M14	M16	1/4-20 UNC	5/16-18 UNC	3/8-16 UNC
P =	1.17	1.48	1.76	2.39	2.97	3.59	4.19	4.88	1.80	2.30	2.78

WSP spotting	Work Material	Vc (m/min)	f (mm/rev.)

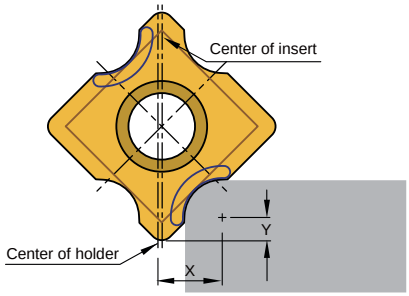


Work Material	Vc (m/min)	f (mm/rev.)
Carbon steel	150 ~ 300	0.05 ~ 0.15
Alloy steel	120 ~ 250	0.05 ~ 0.10
Stainless steel	80 ~ 150	0.04 ~ 0.08
Casting iron	100 ~ 200	0.05 ~ 0.10

N9MT-RC Insert >> Corner Rounding

Determine spindle speed and feed:
To decide running speed of the tools and feed rate, please calculate spindle speed and feed rate according to the following formula and cutting data:

Corner Rounding	Calculate spindle speed



d = 2 x X mm

$$S = \frac{Vc \times 1000}{d \times \pi} \text{ r.p.m.}$$

F = S x f mm/min.

d =	diameter of the tool for calculation purpose
X =	tool radius offset (ref. page 1-18~20 for RC inserts)
Vc =	Cutting Speed -m/min.
S =	Spindle Speed -r.p.m.
F =	mm/min.
f =	mm/rev.

Calculate tool length offest on machining center

TL = TL'-Y,
H = X

X =	tool radius offset (ref. page 1-18~20 for RC inserts)
Y =	distance to the center of radius. (ref. page 1-18~20 for RC inserts)
TL' =	tool length
TL =	tool length offset.
H =	tool radius offset

RC Insert	Work Material	Vc (m/min)	f (mm/rev.)	Grade of Insert



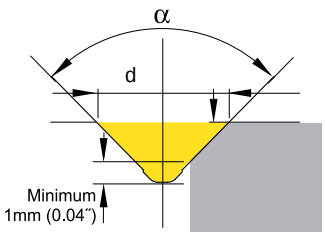
Work Material	Vc (m/min)	f (mm/rev.)	Grade of Insert
Carbon steel	150~320	0.05~0.10	NC40, NC2071
Alloy steel	100~250	0.05~0.10	NC40, NC2071
High alloy steel	80~150	0.04~0.08	NC40, NC2071
Stainless steel	65~125	0.05~0.10	NC9036
Casting iron	150~250	0.05~0.10	NC40, NC2071
Aluminum, Al-alloy Si < 12%	150~320	0.05~0.10	NC9036
Al-alloy Si >12%	100~300	0.05~0.10	NC9036
Cu	200~250	0.05~0.10	NC9036
Brass and Bronze	150~250	0.05~0.10	NC9036
Ti, Ti-alloy	40-80	0.03~0.08	NC9036

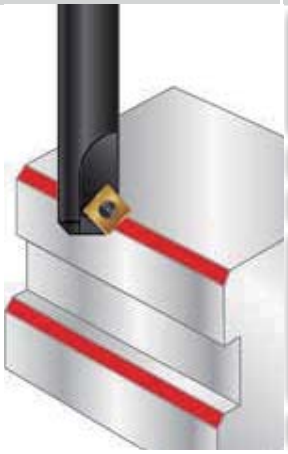
Cutting Data

N9MT-R Insert >> Corner Rounding (4 cutting edges)

R Insert	Work Material	Vc (m/min)	f (mm/rev.)	Grade of Insert
	Carbon steel	150~320	0.05~0.10	NC2071
	Alloy steel	100~250	0.04~0.08	NC2071
	High alloy steel	60~80	0.03~0.06	NC2071
	Casting iron	150~250	0.05~0.10	NC2071

LA Insert >> 45° Chamfering

45° Chamfering	Formula	
	$S = \frac{Vc \times 1000}{d \times \pi}$ r.p.m.	**α =** point angle 90°
		d = effective diameter
		Vc = cutting speed-m/min.or ft/min.
	F = S x f mm/min.	**S =** Spindle speed
		f = feed per rev.-mm/rev.

45° Chamfering	Work Material	Vc (m/min)	f (mm/rev.)	Grade of Insert
	Carbon steel	150-320	0.05~0.10	NC40
	Alloy steel	100-250	0.04~0.08	NC40
	High alloy steel	60-80	0.03~0.06	NC40
	Stainless steel	65-125	0.03~0.06	NC10
	Casting iron	150-250	0.05~0.10	NC10, NC40
	Aluminum, Al-alloy Si < 12%	150-320	0.05~0.10	NC10
	Al-alloy Si >12%	100-300	0.05~0.10	NC10
	Cu	200-250	0.05~0.10	NC10
	Brass and Bronze	150-250	0.05~0.10	NC10
	Hardened steel 40~56 HRC	60-80	0.05~0.10	NC60